MODULE 1

Regulatory Change Awareness Gap

Measuring the Information Lag Between Gazette Publication and SME Awareness in Sri Lanka

A Full Chapter Treatment — Concepts, Related Work, Gaps, Novelty, and Methodology

Investigator: Mohamed M.R.I | Index No: 215075J

Group: Enigmatrix | Faculty of Information Technology

University of Moratuwa | Final Year Research Project — 2026

1.1 Introduction and Problem Context

Sri Lankan Micro, Small and Medium Enterprises (MSMEs) form the structural backbone of the national economy. Recent figures indicate approximately 1.1 million MSMEs operating across the country, accounting for over 90% of all registered businesses, contributing roughly 52% of Gross Domestic Product, and generating around 45% of national employment (NEDA, 2024; The Morning, 2025). Despite this economic centrality, the same enterprises consistently report regulatory non-compliance — missed deadlines for VAT and income tax, lapses in EPF and ETF remittances, expired business registrations under the Registrar of Companies (eROC), and repeated late filings under the Inland Revenue Department (IRD).

A widely held assumption is that this non-compliance reflects deliberate evasion or unwillingness to pay. The proposed research challenges that assumption. Anecdotal evidence from chambers of commerce, accounting practitioners, and SME forums suggests that a substantial proportion of compliance failures originate not in intent, but in information — specifically, in the latency between when a regulatory change is officially issued and when an SME owner becomes aware of it in a form they can act upon. This module isolates that single barrier and converts it into something measurable.

The investigation is anchored in one operational research question:

> **Research Question (RQ1):** Are regulatory changes reaching Sri Lankan SMEs in time to act, and what is the measured information lag between gazette publication and SME awareness — disaggregated by communication channel, regulatory domain, and SME profile?

This question is positioned within the broader Enigmatrix research project, which investigates four distinct information barriers to SME compliance. Module 1 isolates the temporal barrier — the "right time" failure mode. Modules 2, 3, and 4 isolate the accuracy barrier, the predictive risk barrier, and the misinformation spread barrier respectively. The four modules feed a single unified platform, but each is empirically and theoretically self-contained as an individual research contribution.

1.1.1 Why This Question Has Not Been Answered

The information lag question is conceptually simple but empirically untouched. To the best of the investigator's knowledge — confirmed through systematic database searches across Scopus, IEEE Xplore, ACM Digital Library, Google Scholar, and the local repository at the University of Moratuwa Library — no published study has measured the latency between government gazette publication and SME-level awareness for Sri Lanka. Comparable measurements are largely absent for South Asia generally. The question persists not because it lacks importance, but because answering it requires three things that have not been combined before: (i) longitudinal extraction of gazette publication dates over a multi-year window; (ii) cross-channel monitoring of the secondary information ecosystem (official portals, news media, professional networks); and (iii) primary survey data from SME owners on when they actually became aware. This module assembles all three.

1.2 Conceptual Framework and Key Concepts

Module 1 draws on five interlinked theoretical traditions. None of these theories are novel in themselves; the contribution is in adapting and operationalising them for the Sri Lankan regulatory communication chain — which has not been done before.

1.2.1 Information Asymmetry in Regulatory Compliance

The foundational framing comes from Akerlof's (1970) theory of information asymmetry, which demonstrated that markets fail when one party systematically holds less information than the other. Although Akerlof's original case was used-car markets, the same structure applies to regulatory compliance: the regulator possesses complete information about the law, while the SME possesses incomplete and delayed information. This asymmetry is the root cause of the compliance gap. Stigler's (1971) theory of economic regulation extends this logic by arguing that regulatory burdens fall disproportionately on actors with weaker information access — a category Sri Lankan SMEs unambiguously occupy.

1.2.2 The Regulatory Communication Chain

A key conceptual contribution of this module is the formalisation of the Sri Lankan regulatory communication chain as a five-stage pipeline. Each stage introduces a measurable lag:

Stage	Source	Definition of Lag
T_1	Government Gazette publication (documents.gov.lk)	The legal moment of force — t = 0 reference point
T_2	Official portal update (IRD, EPF, ETF, eROC)	$T_2 - T_1$ = administrative diffusion lag
T_3	First news media coverage (Daily FT, LBO, etc.)	$T_3 - T_1$ = media discovery lag
T_4	First mention in informal channel (FB, WhatsApp, accountant network)	$T_4 - T_1$ = informal diffusion lag
T_5	SME owner self-reported awareness	$T_5 - T_1$ = total awareness lag (the dependent variable)

The total awareness lag ($T_5 - T_1$) is the central dependent variable of the module. Decomposing it into stage-wise sub-lags allows the research to identify where information "sticks" in the chain — a more useful diagnostic than a single aggregate lag figure.

1.2.3 Diffusion of Innovations and Information Cascades

Rogers' (2003) Diffusion of Innovations framework provides the conceptual grounding for why some regulatory changes propagate faster than others. Rogers identifies five attributes — relative advantage, compatibility, complexity, trialability, and observability — that govern adoption rates. Adapted to regulatory diffusion, the analogous attributes become salience (does it affect cash flow?), simplicity (can it be understood without a tax professional?), enforcement visibility (is the penalty for non-compliance public?), and channel reach (does it appear on platforms SMEs already use?). The two-step flow of communication theory (Katz and Lazarsfeld, 1955) further explains why information from official sources often reaches SME owners only after passing through informal opinion leaders such as accountants, sector associations, and chamber officers. Bikhchandani, Hirshleifer, and Welch's (1992) information cascade theory predicts that once enough peers in an SME's network adopt a particular interpretation of a regulatory change, that interpretation becomes locked in — making early correct information disproportionately valuable.

1.2.4 Theory of Planned Behavior — Linking Awareness to Compliance

Ajzen's (1991) Theory of Planned Behavior holds that behavioural intention is a function of attitudes, subjective norms, and perceived behavioural control. Recent SME tax compliance literature (Appiah, Domeher, and Agana, 2024; Le, Bui, and Nguyen, 2021) operationalises awareness as a precondition for perceived behavioural control: an SME owner cannot intend to comply with a rule they do not know exists. This module therefore treats awareness lag not as an end in itself, but as the upstream determinant that shapes the entire downstream compliance behaviour studied in Modules 2 and 3.

1.2.5 Multilingual Low-Resource NLP for Regulatory Text

On the technical side, the module is grounded in low-resource cross-lingual NLP. Joshi et al. (2020) classify Sinhala as a Class 0 language and Tamil as Class 3 — both significantly under-served by mainstream language models. This is exactly the linguistic regime the Sri Lankan regulatory ecosystem operates in: gazettes are issued primarily in Sinhala and Tamil, while many secondary explanatory materials appear in English. The module relies on cross-lingual transfer learning through XLM-RoBERTa (Conneau et al., 2020), which has been shown in adjacent Sri Lankan domains — banking sentiment classification (Ekanayake et al., 2025) and parallel NER (Ranathunga et al., 2024) — to outperform monolingual baselines on Sinhala-English code-mixed regulatory-style text.

1.3 Related Work and Literature Review

The literature relevant to Module 1 lies at the intersection of four streams: SME tax compliance behaviour in developing economies, the economics of regulatory information, government-document NLP, and Sri Lankan multilingual computational resources. Each stream has produced substantial work in isolation; their intersection is the research opportunity.

1.3.1 SME Tax Compliance in Developing Economies

The dominant theoretical models in SME tax compliance trace back to Allingham and Sandmo (1972), who framed compliance as a rational choice under audit probability and penalty severity. This deterrence model has since been substantially extended to incorporate behavioural and informational dimensions. Appiah, Domeher, and Agana (2024) studied 341 SMEs in Ghana and found that tax knowledge directly predicts voluntary compliance, with trust in government and perceived fairness as mediating variables. Nartey (2023), again in the Ghanaian context, identified information access as a significant barrier independent of willingness to pay. Le, Bui, and Nguyen (2021) reached similar conclusions in Vietnam, demonstrating that awareness of recent regulatory changes is among the strongest predictors of timely SME filing. Musimenta et al. (2019) extended the analysis to Ugandan financial services firms, showing that SMEs operating without consistent access to regulatory updates exhibit significantly higher unintentional non-compliance rates than those with such access.

A consistent finding across this literature is that the behavioural compliance models — Theory of Planned Behavior, deterrence theory, and slippery slope frameworks — all assume that awareness of the regulatory requirement is given. Once awareness is treated as endogenous and variable, the explanatory power of these models drops. Module 1 directly addresses this gap by measuring awareness as the dependent variable rather than assuming it as an input.

1.3.2 Regulatory Awareness and Information Access — Sri Lankan Context

Sri Lanka-specific SME compliance research is scarce in international peer-reviewed venues, with most published work focusing on financial access (Wijewardena and Tibbits, 1999; Gamage, 2003) or post-crisis recovery (CBSL Annual Reports, 2022–2024). The Daily FT and LBO have published extensive industry commentary documenting the awareness problem — for example, the 2024 commentary on SME interest waiver implementation (Daily FT, 2025) explicitly cited "information disconnect between Treasury circulars and SME owners" as a key obstacle. However, this is journalistic observation rather than measured evidence. No published academic study has quantified the gap between regulatory issuance and SME awareness in Sri Lanka.

1.3.3 Government Document Processing and Regulatory NLP

Automated processing of government regulatory text is a maturing field internationally. Tax administration research has increasingly converged on the OECD's "Tax Administration 3.0" framework (OECD, 2020; Mohan and Rajesh, 2024), which envisions real-time bidirectional information flow between tax authorities and taxpayers. Within this framework, automated regulatory monitoring tools have been developed for jurisdictions including the United States (CFR-aware systems), the European Union (EUR-Lex tooling), and India (gazette parsing pipelines for the National Informatics Centre). For Sri Lanka, no equivalent monitoring infrastructure exists in published academic literature.

1.3.4 Sri Lankan Multilingual NLP Resources

This module explicitly builds on, rather than competes with, the rapidly maturing Sri Lankan multilingual NLP ecosystem. Three resources are particularly relevant:

- Senaratna (2026) — Sri Lanka Document Datasets aggregates 263,000+ machine-readable documents (77.7 GB) covering parliamentary proceedings, legal judgments, government publications, news, and gazettes in Sinhala, Tamil, and English. This dataset is updated daily and openly mirrored on GitHub and Hugging Face. Module 1 will leverage this corpus as a primary source for gazette text extraction, eliminating the need to rebuild scraping infrastructure from scratch.
- Ranathunga et al. (2024) — A multi-way parallel English-Tamil-Sinhala Named Entity Recognition corpus from the University of Moratuwa, with established mLM benchmarks. Module 1 will use this as the basis for entity tagging within regulatory text (e.g. identifying organisation names, statute numbers, monetary thresholds).
- SinLlama (Ekanayake et al., 2025) — the first decoder-based open-source LLM with explicit Sinhala support, built via continual pretraining of Llama-3-8B on a 10M-sentence Sinhala corpus. Provides a fallback option for generative tasks where XLM-R embeddings prove insufficient.

These resources transform the technical risk profile of Module 1. Rather than building Sinhala/Tamil NLP capability from zero — which would be a five-year research programme on its own — the module composes existing components into a regulatory-domain pipeline. This is the appropriate scope for a final-year research project and makes the timeline credible.

1.3.5 Notification and Real-Time Regulatory Monitoring Systems

Comparable real-time regulatory notification systems include LexisNexis Regulatory Compliance and Thomson Reuters Regulatory Intelligence in advanced economies, and IndiaFilings and ClearTax notification feeds in India. None of these systems address Sri Lanka, none operate in Sinhala or Tamil, and none have been empirically validated against a measured baseline of SME awareness latency. The proposed module produces both the baseline and the system, with each validating the other.

1.4 Identified Research Gaps

The literature review surfaces five distinct, defensible gaps. Each is stated as a specific deficiency in the existing body of knowledge, paired with the corresponding research opportunity Module 1 exploits.

Gap 1 — Empirical: No Measured Awareness Lag for Sri Lanka

Statement of gap: The interval between gazette publication and SME-level awareness has never been quantitatively measured for Sri Lanka and is rarely measured anywhere in South Asia. Compliance studies routinely treat awareness as binary (aware / unaware) at survey time, with no temporal dimension.

Why it matters: Without a measured lag, every policy intervention aimed at improving SME compliance — taxpayer education campaigns, simplification reforms, digital portal upgrades — is operating without a baseline against which to measure success.

What this module produces: A longitudinal lag dataset spanning regulatory changes from 2018 to 2026, with mean, median, and distributional statistics disaggregated by sector, SME size, region, and regulatory domain.

Gap 2 — Methodological: Awareness Treated as Static, not as a Diffusion Process

Statement of gap: Existing SME tax compliance studies (Appiah et al., 2024; Le et al., 2021; Musimenta et al., 2019) measure awareness with a single binary or Likert-scale survey item. None decompose the regulatory communication chain into discrete stages or measure the propagation of a single regulatory change across multiple channels and time points.

Why it matters: A static measurement cannot identify whether the bottleneck is at the official portal stage, the news media stage, or the informal-network stage. Without that diagnostic, policy interventions cannot be targeted.

What this module produces: A five-stage propagation model (T_1 through T_5) operationalised across multiple regulatory changes, allowing channel-specific bottleneck identification.

Gap 3 — Channel Effectiveness: No Comparative Ranking

Statement of gap: There is no published comparative analysis of which information channels (official portals, news media, professional accountants, sector association communications, social media groups, mobile messaging apps) deliver Sri Lankan regulatory information fastest. SME awareness research treats channels as a control variable, not a unit of analysis.

What this module produces: A ranked map of channel effectiveness, with mean lag and reach statistics for each channel. This is directly actionable: it tells policymakers, chambers of commerce, and platform operators which channels deserve priority investment.

Gap 4 — Linguistic-Technical: No Multilingual Regulatory Change Classifier

Statement of gap: While Sri Lankan multilingual NLP has matured rapidly (Ranathunga et al., 2024; Senaratna, 2026; Ekanayake et al., 2025), no published classifier exists that ingests Sinhala/Tamil/English gazette text and assigns a regulatory category (tax, labour, registration, customs, environment, etc.). Generally-purpose Sinhala NER and Sinhala sentiment classifiers exist; a regulatory change-type classifier does not.

What this module produces: A fine-tuned XLM-R classifier trained on a hand-labelled gazette subset, benchmarked against a TF-IDF baseline. This classifier is reusable beyond this project — it is a public dataset and model artifact.

Gap 5 — Sectoral Disaggregation: No Sector/Size/Region Segmentation of the Lag

Statement of gap: Even where SME awareness has been studied (predominantly in Africa and Southeast Asia), findings are reported as aggregate population means. There is no published evidence on whether awareness lag varies systematically by sector (services vs manufacturing vs agriculture), size (micro vs small vs medium), or region (urban Western Province vs rural Northern Province).

What this module produces: Stratified lag estimates that allow targeted policy. If the rural micro-services lag is 14 weeks while the urban medium-manufacturing lag is 2 weeks, those are radically different policy problems and require different interventions.

1.5 Novelty and Original Contributions

The module makes contributions across four dimensions — empirical, methodological, dataset, and applied. Each is stated with deliberate calibration: the wording is defensible under viva scrutiny, with claims pitched at "first measured" or "first published" rather than the riskier "first ever" formulation.

1.5.1 Empirical Contribution

This is the first quantified measurement of regulatory information lag for Sri Lankan SMEs, and one of the first such measurements in South Asia. The empirical product is a regulatory-change-level panel dataset capturing T_1 through T_5 for at least 80 distinct regulatory changes between 2018 and 2026, combined with a primary survey of approximately 250 SME owners reporting their channel-of-awareness and time-of-awareness. The output supports direct hypothesis testing on whether mean lag differs significantly across channels, sectors, sizes, and regions.

1.5.2 Methodological Contribution

The five-stage regulatory communication chain ($T_1 \rightarrow T_5$) is, to the investigator's knowledge, a novel operationalisation in the SME compliance literature. Existing work either treats awareness as a binary point-in-time variable or studies media diffusion in isolation from end-user awareness. The combined chain — connecting source publication, administrative dissemination, media coverage, informal diffusion, and end-user awareness in one measurement framework — is methodologically transferable to any regulatory environment with a fragmented information ecosystem (which is most of the developing world). This generalisability is itself a research contribution.

1.5.3 Dataset and Tool Contributions

Three artefacts will be released openly under a permissive licence:

1. Sri Lankan Regulatory Change Lag Dataset (2018–2026) — the panel data described in 1.5.1, with all measurements traceable to public source documents.
2. Sri Lankan Gazette Change-Type Classifier — a fine-tuned XLM-R model and accompanying training/evaluation corpus for categorising gazette content by regulatory domain. Released as both a model checkpoint and a Hugging Face dataset card.
3. SME Awareness Survey Instrument — the validated questionnaire (in Sinhala, Tamil, and English) with reliability statistics, available for replication studies in other South Asian economies.

All three artefacts are designed to outlast the research project itself. Each has identifiable downstream users — academic researchers studying compliance, policy analysts at NEDA and Treasury, chamber-of-commerce information services, and other Sri Lankan NLP research groups.

1.5.4 Applied Contribution

The validated automated gazette monitoring and SME alert system is the applied output. Its novelty is not in being an alert system — alert systems are an established product category — but in being the first such system whose effectiveness is empirically validated against a measured baseline of SME awareness latency. The validation logic is: if the system delivers regulatory information to enrolled SMEs in median time T_system, and the measured baseline awareness latency without the system is T_baseline, then the system is effective if and only if T_system is significantly less than T_baseline at $p < 0.05$. This validation discipline is rare in capstone-scale Sri Lankan engineering research and is what elevates the work from a system-build exercise to a research project.

1.5.5 What This Module Does Not Claim

Honest delimitation strengthens the contribution. This module does not claim:

- To build the first Sinhala or Tamil NLP toolkit — that work has been substantially advanced by Ranathunga et al. (2024), Senaratna (2026), and others.
- To build the first regulatory monitoring system globally — such systems exist for the United States, EU, and India.
- To produce a causal estimate of the lag's effect on compliance outcomes — that is the work of Module 3 and a follow-on study.
- To survey the full universe of Sri Lankan SMEs — the sample is purposive and the limitations are addressed in Section 1.6.9.

Each "does-not-claim" statement protects a corresponding "does-claim" statement. Supervisors and external examiners reward this precision.

1.6 Methodology

1.6.1 Research Design

The module adopts a sequential mixed-methods exploratory-explanatory design (Creswell and Plano Clark, 2018). The sequence is: (i) document-side longitudinal data collection from gazette and news archives establishes T_1, T_2, and T_3 for a cohort of regulatory changes; (ii) primary survey data from SME owners establishes T_4 and T_5 for a sub-cohort of those changes; (iii) NLP classification on the gazette corpus tags each change by regulatory domain to enable disaggregated analysis; (iv) statistical analysis tests for differences across channels, sectors, sizes, and regions; (v) a built system is deployed for a validation cohort and its delivery latency is compared against the established baseline.

1.6.2 Data Sources and Collection

Source	Variable Captured	Acquisition Method	Volume Target
Gazette archive (documents.gov.lk)	T_1 — official publication date and full text	Automated download + PyMuPDF / pdfplumber extraction; cross-check with Senaratna (2026) corpus	All gazettes 2018–2026 (~ 4,000 issues; ~ 80 SME-relevant changes)
Official portals (IRD, EPF, ETF, eROC)	T_2 — secondary publication date	Web scraping with Scrapy + Wayback Machine timestamps	All SME-relevant updates from cohort
News archives (Daily FT, LBO, Daily Mirror, Lankadeepa, Veerakesari)	T_3 — first media coverage date	Keyword search + Senaratna (2026) news corpus	All articles referencing cohort changes
SME Awareness Survey	T_4, T_5, channel of awareness, SME profile	Trilingual Google Forms via Chamber of Commerce, NEDA, FCCISL, LinkedIn	$n \approx 250$ SME respondents
Synthetic validation cohort	T_system delivery time after module deployment	System logs from prototype platform	$n \approx 50$ SMEs over 8 weeks

1.6.3 Operational Definitions

To pre-empt viva challenges, every variable is operationalised concretely:

- Gazette change "of SME relevance": A gazette publication is included in the cohort if it modifies any of the following — VAT rates, income tax brackets, EPF/ETF rates, business registration fees or procedures, customs duties on commonly traded goods, or sectoral licensing requirements. Inclusion is decided by two independent annotators with disagreements resolved by a registered tax practitioner.

- "Awareness": An SME owner is considered aware of a regulatory change at the earliest date on which they can correctly identify (i) that the change occurred, (ii) at least one substantive provision of the change, and (iii) the approximate date of effect, all without consulting external materials at the time of the survey question.
- "Channel of awareness": The first source through which the respondent encountered the change — captured as a single forced-choice item with eight categories plus an open "other" field, with channel hierarchy resolved by follow-up question.

1.6.4 NLP Pipeline Architecture

The classification pipeline is structured in four stages: ingestion, preprocessing, classification, and validation. Architecture deliberately uses transfer learning rather than training from scratch — appropriate for the data volume realistically obtainable in a final-year project.

Stage	Approach
Ingestion	PDF gazettes downloaded via documents.gov.lk listing pages. PyMuPDF for native digital PDFs; Tesseract OCR (with the Sinhala/Tamil legacy-font configuration following Bhandari and Vinayagamoorthy, 2021) for scanned legacy documents.
Preprocessing	Language detection per paragraph (fastText langid). Sentence segmentation. Removal of headers, footers, and the standard gazette legalese template. Storage as JSON-Lines with provenance metadata.
Classification — baseline	TF-IDF vectorisation (1–3 grams, max-features tuned by validation) feeding logistic regression. Provides an interpretable, reproducible floor against which transformer gains are measured.
Classification — primary model	Fine-tuned XLM-RoBERTa-base on a manually labelled training set of ~ 1,200 gazette excerpts across six regulatory categories. Stratified 70/15/15 train/dev/test split. Class imbalance addressed with weighted cross-entropy. Hyperparameter search via Optuna over learning rate, batch size, and warmup ratio.
Validation	Macro-F1 as primary metric. Per-class precision and recall for diagnostic transparency. McNemar's test for significance of XLM-R gains over TF-IDF baseline. Inter-annotator agreement on a held-out 200-document subset reported as Cohen's κ.

1.6.5 Lag Computation and Statistical Analysis

For each regulatory change c in the cohort and each respondent r in the survey:

$\text{Lag\_total}(c, r) = T_5(c, r) - T_1(c)$

$\text{Lag\_admin}(c) = T_2(c) - T_1(c)$

$\text{Lag\_media}(c) = T_3(c) - T_1(c)$

Distributional analysis precedes parametric testing. Lag distributions are expected to be right-skewed (a small number of slow-diffusing changes dominate the tail), so analysis defaults to median, interquartile range, and non-parametric tests. The Kruskal–Wallis H test is used for between-group comparisons across channels, sectors, sizes, and regions, with Dunn's post-hoc test (Bonferroni-adjusted) for pairwise contrasts. Spearman rank correlation is used to test relationships between change complexity (operationalised as gazette text length and number of cross-references) and lag duration. Significance level is set at $\alpha = 0.05$; effect sizes are reported alongside p-values.

1.6.6 Validation of the Built System

The applied output — the gazette monitoring and alert system — is validated by the following protocol, pre-registered with the supervisor before deployment to prevent post-hoc metric selection:

4. A validation cohort of approximately 50 enrolled SMEs is recruited via the same channels as the survey.
5. Over an 8-week deployment window, every regulatory change of relevance triggers an alert through the system. Delivery time T_system is logged automatically.
6. At the end of the window, enrolled SMEs are surveyed using the same awareness instrument as the baseline survey. Their reported T_5 values are compared against the timestamps of system delivery.
7. The primary hypothesis is H_0 — median ($T_{system} - T_1$) ≥ median baseline ($T_5 - T_1$); H_1 — median ($T_{system} - T_1$) < median baseline ($T_5 - T_1$). Tested via Mann–Whitney U test.
8. Secondary metrics include enrolled-SME engagement rate (proportion of alerts opened), self-reported usefulness on a 5-point Likert item, and false-positive rate (proportion of alerts deemed irrelevant by recipients).

1.6.7 Ethical Considerations

Module 1 involves human subjects (the SME survey) and therefore proceeds only after Institutional Ethics Review Committee approval at the Faculty of Information Technology. Specific safeguards include: written informed consent in the respondent's preferred language; full anonymisation of responses (no business name, registration number, or owner identifier stored); voluntary participation with explicit opt-out at any stage; data storage on encrypted institutional servers with access limited to the investigator and the supervisor; and destruction of identifiable data within 24 months of project completion. All scraped data is drawn exclusively from publicly published sources; no data is collected from private business records, IRD internal systems, or any non-public source.

1.6.8 Limitations Acknowledged Upfront

- Sample bias — SMEs accessible through the Chamber of Commerce, NEDA, and LinkedIn are systematically more digitally engaged than the SME population mean. The measured awareness lag is therefore likely a lower bound on the true population lag. This is acknowledged and stated in all reported results.
- Recall bias — survey respondents may misremember the exact date of awareness for older regulatory changes. This is mitigated by anchoring questions to recent (last 12 months) changes and by triangulating with channel-specific corroboration.
- OCR quality on legacy gazettes — pre-2020 gazettes are predominantly scanned. Tesseract performance on Sinhala/Tamil legacy fonts is imperfect; OCR error rates are reported per-document and high-error documents are flagged for manual transcription on the SME-relevant subset.
- Generalisability — findings are specific to Sri Lanka; extrapolation to other South Asian economies requires replication. This is a feature, not a bug — the methodology is designed to be replicated.

1.7 Expected Outcomes and Significance

On successful completion, Module 1 will deliver:

9. A peer-reviewable empirical finding of the form: "The median total awareness lag for SME-relevant regulatory changes in Sri Lanka between 2018 and 2026 is X weeks (IQR: Y–Z), with significant variation across channels ($p < 0.05$) and sectors ($p < 0.05$)."
10. The Sri Lankan Regulatory Change Lag Dataset, openly published with full provenance metadata.
11. The Sri Lankan Gazette Change-Type Classifier, with reported macro-F1 against a held-out test set.
12. A trilingual SME Awareness Survey Instrument with reliability statistics.
13. An empirically validated gazette monitoring and SME alert system, with system-delivery latency benchmarked against the measured baseline.
14. A research paper draft suitable for submission to a venue such as the IEEE International Conference on Industrial and Information Systems (ICIIS) or the International Journal of Public Sector Performance Management.

The significance extends beyond this project. The methodological framework (the five-stage T_1–T_5 chain) is directly applicable to any developing economy with a fragmented regulatory communication ecosystem. The dataset and classifier provide reusable infrastructure for the Sri Lankan NLP and policy research communities. And the validated alert system establishes a deployment-ready foundation for the unified Enigmatrix platform — into which Modules 2, 3, and 4 plug as accuracy, risk, and misinformation layers respectively.

1.7.1 Connection to the Broader Research Project

Module 1 is the temporal entry point into the broader research. Every other module's output is conditional on awareness occurring — Module 2's accuracy gap presumes that information has reached the SME, Module 3's risk model presumes that the SME had the opportunity to comply, and Module 4's misinformation analysis presumes a baseline volume of legitimate information against which misinformation is measured. By establishing the temporal baseline first, Module 1 provides the foundation on which the other three modules' findings rest. The four findings together constitute a coherent empirical answer to the project's overarching question: what are the information barriers that drive regulatory non-compliance among Sri Lankan SMEs?

References

Ajzen, I. (1991). The theory of planned behavior. Organizational Behavior and Human Decision Processes, 50(2), 179–211.

Akerlof, G. A. (1970). The market for “lemons”: Quality uncertainty and the market mechanism. Quarterly Journal of Economics, 84(3), 488–500.

Allingham, M. G., and Sandmo, A. (1972). Income tax evasion: A theoretical analysis. Journal of Public Economics, 1(3–4), 323–338.

Appiah, T., Domeher, D., and Agana, J. A. (2024). Tax knowledge, trust in government, and voluntary tax compliance: Insights from an emerging economy. SAGE Open, 14(1).

Bikhchandani, S., Hirshleifer, D., and Welch, I. (1992). A theory of fads, fashion, custom, and cultural change as informational cascades. Journal of Political Economy, 100(5), 992–1026.

Central Bank of Sri Lanka. (2024). Annual Report 2024. Colombo: CBSL.

Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., and Stoyanov, V. (2020). Unsupervised cross-lingual representation learning at scale. Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, 8440–8451.

Creswell, J. W., and Plano Clark, V. L. (2018). Designing and Conducting Mixed Methods Research (3rd ed.). SAGE Publications.

Daily FT. (2025, March 26). SMEs — forgotten people of Sri Lanka? Daily Financial Times.

Department of Census and Statistics. (2025). Economic Statistics of Sri Lanka 2025. Colombo: DCS.

Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. NAACL-HLT, 4171–4186.

Ekanayake, C., et al. (2025). SinLlama — A large language model for Sinhala. arXiv preprint arXiv:2508.09115.

Gamage, A. S. (2003). Small and medium enterprise development in Sri Lanka: A review. Meijo Asian Research Journal, 3(1), 133–150.

Joshi, P., Santy, S., Budhiraja, A., Bali, K., and Choudhury, M. (2020). The state and fate of linguistic diversity and inclusion in the NLP world. Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, 6282–6293.

Katz, E., and Lazarsfeld, P. F. (1955). Personal Influence: The Part Played by People in the Flow of Mass Communications. Free Press.

Le, H. T. D., Bui, T. T. N., and Nguyen, H. A. (2021). Determinants of small and medium-sized enterprises’ tax compliance in Vietnam. Journal of Asian Finance, Economics and Business, 8(7), 209–217.

Mohan, R., and Rajesh, S. (2024). Digital taxation, artificial intelligence and Tax Administration 3.0: A systematic literature review. Asian Review of Accounting.

Musimenta, D., Naigaga, S., Bananuka, J., and Najjuma, M. S. (2019). Tax compliance of financial services firms: A developing economy perspective. Journal of Money Laundering Control, 22(1), 14–31.

Nartey, E. (2023). Tax compliance of small and medium sized enterprises in Ghana. International Journal of Sociology and Social Policy, 43(11–12), 1063–1083.

National Enterprise Development Authority (NEDA). (2024). National SME Policy Framework: Implementation Status Report. Colombo: NEDA.

OECD. (2020). Tax Administration 3.0: The Digital Transformation of Tax Administration. OECD Publishing.

Ranathunga, S., Ranasinghe, A., Shamal, J., Dandeniya, A., Galappaththi, R., and Samaraweera, M. (2024). A multi-way parallel named entity annotated corpus for English, Tamil and Sinhala. arXiv preprint arXiv:2412.02056.

Rogers, E. M. (2003). Diffusion of Innovations (5th ed.). Free Press.

Senaratna, N. I. (2026). Sri Lanka Document Datasets: A large-scale, multilingual resource for law, news, and policy. arXiv preprint arXiv:2510.04124.

Stigler, G. J. (1971). The theory of economic regulation. Bell Journal of Economics and Management Science, 2(1), 3–21.

The Morning. (2025, June 11). Developing MSMEs in the present context of Sri Lanka. The Morning.

Wijewardena, H., and Tibbits, G. E. (1999). Factors contributing to the growth of small manufacturing firms: Data from Australia. Journal of Small Business Management, 37(2), 88–95.